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28 May 1987
SUPERSEDING
MIL-D-43699B
18 July 1984

MILITARY SPECIFICATION

DRUMS, FABRIC-REINFORCED, COLLAPSIBLE: (DRINKING WATER; 55-, 250- and 500 GALLON CAPACITIES)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of non-vented, cylindrical, collapsible, fabric-reinforced elastomer drum in three sizes.

1.2 Classification. The drums covered by this specification shall be the following sizes:

Size I - 250 gallon capacity.
Size II - 500 gallon capacity.
Size IIA - 500 gallon capacity (with enlarged [4 inch] filler/discharge).
Size III - 55 gallon capacity.

1.2.1 Definitive specification part number. The specification part number is a definitive part number which will be formulated to identify each item covered by this specification. The part number will be formulated by selecting from the requirement options available in this specification as follows:

Definitive specification part number:

M43699 - XX
- XX: Size designator (see 1.2.2)
- M43699: Specification number (without revision letter)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.2 Size designator. The size designator is a two position field used to designate the required drum capacity:

01 - 250 gallon
02 - 500 gallon
03 - 55 gallon
04 - 500 gallon (4 inch filler discharge)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-W-461	- Wire, Steel, Carbon (Round, Bare and Coated).
ZZ-H-601	- Hose and Hose Assemblies, Rubber (Yarn or Fabric Reinforced) Water Service.
PPP-B-35	- Bags, Textile, Shipping, Burlap, Cotton and Waterproof Laminated.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-C-795	- Cushioning Material, Packaging (Flexible Cellular Plastic Film) for Packaging Applications.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-C-1797	- Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam.
PPP-P-291	- Paperboard, Wrapping and Cushioning.
PPP-T-60	- Tape, Packaging, Waterproof.
PPP-T-97	- Tape, Pressure Sensitive Adhesive, Packaging/ Paper.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-6396	- Tank, Fuel, Oil, Water-Alcohol, Coolant Fluid, Aircraft, Non-Self Sealing, Removable, Internal.
MIL-S-7916	- Sealing Compound, Thread and Gasket, Fuel, Oil and Water Resistant.
MIL-A-8625	- Anodic Coatings, for Aluminum Alloys.

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MIL-L-10547 - Liner, Case, and Sheet, Overwrap, Water Vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL

FED-STD-595 - Color.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing with Appropriate Test Methods.
MS39352 - Adapter, Male by Male, Quick Disconnect, Cam-Locking Type, Type XXI.
MS49002 - Reducer, External Pipe Thread oy Quick Disconnect Cam-Locking Type, Female, Type XIII.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ME

TA13226E6020 - Drum, Fabric, Collapsible, Drinking Water, 55 Gallons.
TA13226E6030 - Drum, Fabric, Collapsible, Drinking Water, 250 Gallons.
TA13226E6040 - Drum, Fabric, Collapsible, Drinking Water, 500 Gallons.
TA13228E3190 - Drum, Fabric, Collapsible, Drinking Water, 500 Gallons, (4 inch Filler/Discharge).

FOOD AND DRUG ADMINISTRATION (FDA)

Title 21, Code of Federal Regulations, Chapter 1, Part 177.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 153 - Zinc Coating (Hot Dip) on Iron and Steel Hardware.
ASTM A 167 - Plate, Sheet, and Strip, Stainless and Heat Resisting Chromium-Nickel Steel.
ASTM A 366 - Steel Sheets, Carbon, Cold-Rolled, Commercial Quality.
ASTM A 380 - Cleaning and Descaling Stainless Steel Parts Equipment and Systems.
ASTM A 635 - Steel, Sheet and Strip, Heavy-Thickness Coils, Carbon, Hot Rolled.
ASTM D 2000 - Rubber Products in Automotive Applications.
ASTM E 8 - Tension Testing of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN PUBLIC HEALTH ASSOCIATION, INC. (APHA)

Standard Methods for the Examination of Water and Wastewater.

(Application for copies should be addressed to the American Public Health Association, Inc., 1015 18th Street, Washington, DC 20036.)

NATIONAL SANITATION FOUNDATION (NSF)

NSF Standard 14 - Official Plastics Listing.

(Application for copies should be addressed to the National Sanitation Foundation, 3475 Plymouth Rd, PO Box 1468, Ann Arbor, MI 48106.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail

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specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The drinking water drums shall be in accordance with Top Assemblies TA13226E6030 for size I, TA13226E6040 for size II, TA13228E3190 for size IIA, TA13226E6020 for size III, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the drums. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of drums from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish drums conforming to this specification.

3.3 Taste and odor. The drum shall not impart to chlorinated water stored therein an odor of an intensity greater than threshhold odor No. 2, nor a taste of an intensity greater than rating level No. 4 based upon procedures specified in the American Public Health Association (APHA) Standard Methods for the Examination of Water and Wastewater for taste and odor.

3.4 Material. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specifications.

3.4.1 Liner. The liner material and all surfaces that contact drinking water shall conform to the Code of Federal Regulations, Food and Drugs, title 21, chapter 1, parts 177.2600 and title 21, chapter 1 for contact with food, (the compounds listed in title 21, chapter 1, part 177.1020, 177.1030, 177.1050 or 177.1480 shall not be acceptable materials) or the liner material and all surfaces that contact drinking water shall be tested and listed as acceptable for use with drinking water by the National Sanitation Foundation under Standard 14.

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3.4.1.1 Certification. When specified (see 6.2), prior to the approval of the first article or prior to approval of the first shipment, the contractor shall furnish the contracting officer a certificate of compliance that the liner and all surfaces that contact drinking water complies with 3.4.1.

3.4.2 Material deterioration prevention and control. The drums shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the drums may be exposed.

3.4.3 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.4.4 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.4.5 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the drums may be newly fabricated from recovered materials to the maximum extent practicable, provided the drums produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the drums. No recovered materials shall be used for materials which contact drinking water.

3.4.6 Inner liner elastomer compound. The liner elastomer compound shall be a solid, noncellular elastomer compound conforming to ASTM D 2000, type M4AA510A13EA14F17.

3.4.7 Outer cover elastomer compound. The outer cover elastomer compound shall conform to ASTM D 2000, type M3BC610A14EF21C12F17. Maximum swell requirement for EF21 shall be 66 percent.

3.4.8 Cord or fabric elastomer compound. The cord fabric elastomer compound shall conform to ASTM D 2000, type M2AA510F17.

3.4.9 Steel, corrosion-resisting. Corrosion-resisting steel shall conform to ASTM A 167, UNS S30400, passivated in accordance with ASTM A 380.

3.4.10 Compound, thread sealing. All threaded connections of the metal fittings shall be sealed with sealing compound conforming to MIL-S-7916. The sealing compound shall not cause thread seizure or galling.

3.4.11 Gaskets. All valves shall be provided with FDA or NSF approved elastomer gaskets serviceable to 160 °F. When specified (see 6.2), a

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certificate of compliance shall be provided by the contractor that material of the elastomer gaskets serviceable to 160 °F complies with Code of Federal Regulations, Food and Drug or the National Sanitation Foundatiuon Standard 14 (see 3.4.1).

3.5 Construction. The drum shall be cord or fabric reinforced elastomer compound with an inner liner of material specified in 3.4.6 and an outer cover of material as specified in 3.4.7. Where the drum body has been cut for placement of the flanges or seams, any cords exposed to the interior shall be sealed with the elastomer compound specified in 3.4.6.

3.6 Physical characteristics.

3.6.1 Puncture resistance. The drum wall (excluding the reinforced ends) shall withstand a force of not less than 200 pounds without being punctured. The sample referenced in 4.4.2.2 shall be used to determine puncture resistance.

3.6.2 Weight. The weight of the completely assembled drum shall not exceed 240 pounds for size I, 275 pounds for size II, 300 for size IIA, and 50 pounds for size III.

3.7 Performance.

3.7.1 Ultimate pressure. The drum shall withstand an ultimate pressure of 45 pounds force per square inch gage (psig) for size I, II and IIA, 35 psig for size III, without any visual evidence of leakage, external or internal component layer separation, delamination propagation or blistering.

3.7.2 Proof pressure. The drum shall withstand a hydrostatic pressure of 30 psig for size I, II and IIA, 20 psig for size III without any visual evidence of leakage.

3.7.3 Ambient temperature collapsibility. The drum shall be filled and emptied through 75 cycles in an ambient temperature of 60 to 90 °F, without any visual evidence of leakage, external or internal blistering, delamination, splits, cracks, or chipping.

3.7.4 Leakage. The drum shall withstand the leakage test specified in 4.5.2.11 without visual evidence of leakage.

3.8 Finish. After complete fabrication and prior to assembly of the drum, the inside and outside shall be cleaned of all foreign material. The color of the outer surface, except for the blue rings on the drum ends and identification markings, shall be sand matte, approximate FED-STD-595, chip number 30277.

3.9 Marking for identification. Each drum shall be marked as shown on drawings in accordance with MIL-STD-130.

3.10 Detailing of size requirements.

3.10.1 Size I and II.

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3.10.1.1 Expansion dimensions. Drum dimensional changes shall not exceed the following when subjected to a pressure of 30 psig for 7 hours:

Overall length	- 1 inch
Diameter	- 6 inches

3.10.1.2 Sleeve and wire rope assembly. The sleeve and wire rope assembly, shall have a breaking strength of not less than 12,000 pounds.

3.10.1.3 Anodic coating. All aluminum surfaces shall have rough and sharp edges and scale removed. These surfaces shall be cleaned, anodized, and dyed in accordance with MIL-A-8625, type II, class 2, thickness 0.0007 inch for wrought aluminum alloys and 0.0004 inch for aluminum alloy castings. The color shall be nonvivid and nonreflective and shall visually approximate color number 30277 of FED-STD-595.

3.10.1.4 Drop. The drum, when full of water, and in the horizontal position, shall be capable of withstanding five successive drops from a height of not less than 5 feet without any visual evidence of leakage, broken cables or hardware, or separation of component layers of the drum body.

3.10.1.5 Rolling tow. The drum shall be capable of being towed as a wheel for 10 miles over paved road without any visual evidence of leakage.

3.10.2 Size III.

3.10.2.1 Steel, sheet and strap. Steel shall conform to ASTM A 366, cold-rolled sheet, commercial quality, class 1, matte finish or ASTM A 635, grade 1010, hot rolled sheet and strip.

3.10.2.2 Wire, steel. Steel wire shall conform to QQ-W-461, AISI 1010, 1015 or 1020, annealed-in-process.

3.10.2.3 Flange and cable assembly. The cable sleeves, flanges and cable shall conform to drawing A2-13226E6015 and shall have a breaking strength of not less than 5000 pounds.

3.10.2.4 Hardware, drum. The D-ring, top plate, and outer plate shall be cleaned of all foreign matter. After cleaning, the parts shall be galvanized by the hot-dip method in accordance with ASTM A 153.

3.11 Workmanship. Workmanship shall not permit defects adversely affecting the strength or serviceability of the finished drum. All metal parts shall be cleaned and free from sand, dirt, scale, and flux. Surfaces shall be smooth with edges rounded or beveled. The inside and outside of the drum shall be clean and free from foreign materials. Any necessary repair or rework shall restore the reworked area to its full strength and shall meet all applicable requirements of this specification. The drum shall not leak.

3.12 Accessories. Unless otherwise specified (see 6.2) each 250 gallon drum shall be provided with the following accessories:

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Item	Conforming to	Quantity
MS39352-1	Adapter, Male x male, quick disconnect, 1-1/2 x 1-1/2-inch	1
13225E9136-9 (97403)	Hose Assy, per ZZ-H-601, 1-1/2-inch diameter, 10-foot length	1
MS49002-17	Reducer, 1-1/2 x 1 inch, quick disconnect	1
13225E91381-1 (97403)	Swivel, 1 inch, quick disconnect by threaded	1
13226E1707 (97403)	Nozzle, 1 inch (0.812-inch nominal diameter outlet)	1

All accessories shall approximate sand matte, color number 30277 of FED-STD-595. All accessories shall be suitable for use with drinking water.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection failure. Failure of a first article drum to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the first article drums, and shall be cause for refusal by the Government to continue acceptance of production drums until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on drums previously accepted and produced under the contract. Any deficiencies found as a result of the first article inspection will be considered prima facie evidence that all drums accepted prior to the completion of the first article inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.1.2 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies

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submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.3 Disassembly inspection. Failure of any examination or test by the first article drum shall be cause for disassembly, in the presence of a Government representative, of the first article drum to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be the responsibility of the contractor.

4.1.4 Process inspection. Inspection shall be accomplished by the contractor to assure compliance with the approved documented work instructions covering processing and fabricating procedures. Any process or fabricating procedure not in accordance with the approved procedure shall be cause to withhold acceptance of drums determined to have been manufactured using unapproved processing or fabricating procedure.

4.1.5 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3. First article inspection.

4.3.1 Examination. The first article drums shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.3.

4.3.2 Tests. The first article drums and material samples shall be subjected to tests marked "X" in column 1 of table I. Every test specified in 4.5.2 except 4.5.2.7 and 4.5.2.11 shall be conducted on one of two drums so that approximately one-half of the tests shall be performed on each drum. Tests specified in 4.5.2.7 and 4.5.2.11 shall be conducted on each of two drums. Tests specified in 4.5.2.2 and 4.5.2.3 shall be performed on the material samples. The material samples shall be representative of materials used to fabricate the drum body. Failure of any test shall be cause for rejection or for performing the inspection specified in 4.1.3.

4.4 Quality conformance inspection.

4.4.1 Lot. A lot shall consist of not more than 300 drums of the same size, each drum having successfully passed the tests specified in 4.4.4.1.

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4.4.2 Sampling for examination and tests.

4.4.2.1 Drums. Two drums in size I, II and IIA, shall be selected at random from each lot for examination and tests. One drum in size III shall be selected at random from each lot for examination and test.

4.4.2.2 Drum body. One sample of drum body material approximately 1-foot square and of the approximate thickness of the drum body shall be prepared from the elastomer compounds and cord or fabric used in fabricating the drums of the lot. The drum sample shall be cured equivalent to the drums.

4.4.3 Examination. Samples selected in accordance with 4.4.2.1 shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection of the represented lot of drums.

4.4.4 Tests.

4.4.4.1 Individual. Each drum shall be subjected to the tests marked "X" in column 2 of table I. Failure of either test shall be cause for rejection of the drum.

4.4.4.2 Samples.

4.4.4.2.1 Drums. Drums selected in accordance with 4.4.2 shall be subjected to tests marked "X" in column 3 of table I as follows:

a. Size I and II. One of the drums of size I, II and IIA, shall be tested as specified in 4.5.2.5 and 4.5.2.6. The sleeve and wire rope assembly shall be removed from the drum and tested as specified in 4.5.2.13. The remaining drum of size I and II shall be tested as specified in 4.5.2.4 and 4.5.2.8. Failure of any test shall be cause for rejection of the represented lot of drums.
b. Size III. The drum shall be tested as specified in 4.5.2.5 and 4.5.2.6. The sleeve, cable flanges and cable shall be removed from the drum and tested as specified in 4.5.2.12. Failure of any test shall be cause for rejection of the represented lot of drums.

4.4.4.2.2 Drum material samples. Samples selected in accordance with 4.4.2 shall be tested as specified in 4.5.2.3. Failure of test shall be cause for rejection of the represented drums.

4.5 Inspection procedure.

4.5.1 Examination. The drum shall be examined as specified herein for the following defects:

101. Dimensions not as specified.
102. Materials not as specified.
103. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.

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104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
105. Contractor does not have documentation available for identification of material, material finishes, or treatment.
106. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the drum.
107. Inner liner elastomer compound not as specified.
108. Outer cover elastomer compound not as specified.
109. Cord or fabric elastomer compound not as specified.
110. Corrosion-resisting steel not as specified.
111. Thread-sealing compound not as specified.
112. Gaskets not as specified.
113. Construction not as specified.
114. Fittings not as specified.
115. Finish of fittings not as specified.
116. Fittings not located as specified.
117. Finish not as specified.
118. Color not as specified.
119. Identification label and marking not as specified.
120. Anodic coating not as specified.
121. Steel, sheet and strap, not as specified.
122. Steel wire not as specified.
123. Drum hardware not as specified.
124. Components fractured, split, sprung, malformed, or missing.
125. Workmanship not as specified.
126. Certificate of compliance for FDA or NSF compliant liner and gasket material missing, when applicable.
127. Accessories not provided for each 250 gallon drum as specified.

4.5.2 Tests.

4.5.2.1 Test schedule.

4.5.2.2 Taste and odor. A sample of the drum wall material shall be immersed for 70 hours in distilled water having a 0.2 parts per million of total available chlorine at the start of the 70-hour test. The samples shall be large enough to expose 36 square centimeters of compound to one liter of chlorinated distilled water. The material shall not impart odor to chlorinated water to an intensity greater than threshold odor No. 2. The material shall not impart taste to chlorinated water to a rating level higher than 4. The water shall be tested for taste and odor in accordance with procedures outlined in APHA Standard Methods for the Examination of Water and Wastewater. Failure to meet these levels shall constitute failure of this test.

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TABLE I. Test schedule.

First article	Quality conformance: Individual	Quality conformance: Sample	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
X			Taste and odor	4.5.2.2	3.3
X		X	Puncture resistance	4.5.2.3	3.6.1
X		X	Expansion (size I, II and IIA)	4.5.2.4	3.10.1.1
X		X	Weight	4.5.2.5	3.6.2
X		X	Ultimate pressure	4.5.2.6	3.7.1
X	X		Proof pressure	4.5.2.7	3.7.2
X		X	Drop (size I, II and IIA)	4.5.2.8	3.10.1.4
X			Rolling tow (size I, II, and IIA)	4.5.2.9	3.10.1.5
X			Ambient temperature collapsibility	4.5.2.10	3.7.3
X	X		Leakage	4.5.2.11	3.7.4
X		X	Cable assembly pull test (size III)	4.5.2.12	3.10.2.3
X		X	Sleeve and wire rope assembly (size I, II, and IIA)	4.5.2.13	3.10.1.2

4.5.2.3 Puncture resistance. A sample of the drum wall material (excluding the reinforced ends) shall be prepared according to 4.4.2.2. shall be puncture tested as specified in MIL-T-6396, except that the force required shall be 200 pounds. Nonconformance to 3.6.1 shall constitute failure of this test.

4.5.2.4 Expansion (size I, II and IIA)). The drum shall be filled with water to a pressure of 30 psig at an ambient temperature of 65 to 75 °F. The length and diameter shall be measured within 15 minutes of filling. The length shall be measured by placing a 90 degree square upright at each end and measuring the distance between squares. The diameter shall be similarly measured by placing the square upright against each side at the center of the drum and measuring the distance between squares. The measurements shall be the original dimensions. The drum shall be maintained at 30 psig for 7 hours. At the end of the 7 hours, the measurements of length and diameter shall be repeated as described above. The difference between these final measurements and the original measurements shall be regarded as the change in dimensions. Nonconformance to 3.10.1.1 shall constitute failure of this test.

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4.5.2.5 Weight. The emptied drum, with hardware, shall be weighed. Nonconformance to 3.6.2 shall constitute failure of this test.

4.5.2.6 Ultimate pressure. The drum shall be subjected to a hydrostatic pressure of 45 psig for size I, II and IIA, and 35 psig for size III and allowed to stand for 30 minutes to determine compliance with 3.7.1. At the end of 5 minutes, the pressure shall be readjusted to 45 psig, for size I, II and IIA, and 35 psig for size III, if the pressure has dropped due to expansion of the drum. After the hydrostatic test, the drum shall be emptied and visually examined both externally and internally. Nonconformance to 3.7.1 shall constitute failure of this test.

4.5.2.7 Proof pressure. Each drum shall be subjected to hydrostatic pressure of 30 psig for size I, II and IIA, and 20 psig for size III and allowed to stand for 30 minutes. At the end of 5 minutes, the pressure shall be readjusted to 30 psig for size I, II and IIA, and 20 psig for size III if the pressure has dropped due to expansion of the drum. Nonconformance to 3.7.2 shall constitute failure of this test.

4.5.2.8 Drop (size I, II and IIA). After the test in 4.5.2.7 has been completed, the drum shall be drained and refilled with water. For this test "full" means 225 gallons in size I and 475 gallons in size II and IIA. The drum shall then be lifted by the two anchor shackles on the ends until the tank has a minimum ground clearance of 5 feet. The drum shall free fall onto unprepared ground such as grass, sand, or bare earth. After the five free falls, the drum shall be visually examined for leaks, then emptied and visually examined, internally and externally, for evidence of broken hardware and separation of component layers of the drum body. The time between draining the water and completion of the free fall shall not exceed 3 hours. Nonconformance to 3.10.1.4 shall constitute failure of this test.

4.5.2.9 Rolling tow (size I, II and IIA). The drum shall be filled with not less than 225 gallons of water in size I and 475 gallons in size II and IIA pressurized to 5 psi. A suitable towbar shall be attached to the lugs, and the drum shall be towed as a wheel for not less than 10 miles over a paved road. The rate of speed of towing shall be between 5 to 10 miles per hour (mph). Nonconformance to 3.10.1.5 shall constitute failure of this test.

4.5.2.10 Ambient temperature collapsibility. The emptied drum shall be filled with a measured volume of water, then emptied using a suitable pump and subjecting the drum to a vacuum of not less than 12 inches of mercury.

Water volume for collapsibility test

Size	Fill	Remove
I	240 gallons	210 gallons
II and IIA	465 gallons	435 gallons
III	45 gallons	35 gallons

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This test shall be conducted at an ambient temperature of 60 to 90 °F. With the drum in a horizontal position, the drum shall be filled and emptied through 75 cycles, (each filling and emptying constitutes one cycle). The drum shall be disassembled for examination of the drum internal carcass. Nonconformance to 3.7.3 shall constitute failure of this test.

4.5.2.11 Leakage. Each drum shall be inflated with air to 6 ±0.5 psig, and sprayed with soap or detergent sudsing solution over the entire surface to determine compliance with 3.7.4. Leakage shall be determined by observing areas for fizzing or bubbling of soap or detergent sudsing. If any evidence of fizzing or bubbling is observed, a Freon test shall be made on the drum. Inflate drum to 2 psig with air, add Freon to bring pressure to 4 psig and add additional air to bring the drum to 6 psig and allow to stand for 15 minutes. At the end of the 15 minutes, use a Halogen leak detector to determine if the drum is defective. Evidence of leakage determined by the Freon test shall constitute failure of this test.

4.5.2.12 Cable assembly pull test (size III). The sleeve, cable flanges and the cable shall be assembled. A device shall be attached to the inside lip of the cable flanges and pulled to determine the breaking strength. Nonconformance to 3.10.2.3 shall constitute failure of this test.

4.5.2.13 Sleeve and wire rope assembly (size I, II and IIA). The sleeve and wire rope assembly shall be tested in accordance with the applicable requirements of ASTM E 8, to determine the breaking strength. Nonconformance to 3.10.1.2 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

128.	Shackle not wrapped and taped as specified for level A (see 5.1.1.1).
129.	Separate components not individually wrapped, taped, marked and consolidated as specified for level A (see 5.1.1.2).
130.	Technical manual not preserved as specified for level A (see 5.1.1.3).
131.	Items not preserved for level C as specified (see 5.1.2).
132.	Boxes not as specified for all levels (see 5.2.1, 5.2.2, and 5.2.3).
133.	All items not blocked, cushioned and braced as specified for all levels (see 5.2.1, 5.2.2, and 5.2.3).
134.	Box closure and strapping not as specified for all levels, (see 5.2.1, 5.2.2 and 5.2.3).
135.	Marking missing, illegible, incorrect or incomplete for all levels (see 5.2.4).

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Drum. Each drum shall be throughly drained and the interior and exterior shall be dried prior to packaging. Each shackle shall be wrapped with corrugated, single faced paperboard material conforming to PPP-P-291, type I, style 1, which shall be secured with tape conforming to PPP-T-60, or PPP-T-97, type IV. Each drum shall be collapsed and compactly folded so that one swivel plate is on the interior of the bundle and the other is on the top exterior of the bundle.

5.1.1.2 Separate components. The valves and adapters, nozzles, hoses, quick disconnect adapters and other loose items shall be wrapped individually with cushioning material conforming to PPP-C-795, PPP-C-1797, or PPP-C-843 (type II, class B). The cushioning material shall be secured in place with tape conforming to PPP-T-60 or PPP-T-97 (type IV). Each individually wrapped item shall be marked properly for identification purposes. All components shall be consolidated in a bag conforming to PPP-B-35 (type I or II).

5.1.1.3 Technical manual. The technical manual shall be preserved in accordance with MIL-116, method IC-3.

5.1.2 Level C. Each drum, separate components and technical manual shall be preserved individually in accordance with MIL-P-116, method III.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Each complete drum, separate components, and technical manual, preserved as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-621, class 2, style 2 or 2-1/2, type 3 load. Each box shall be equipped with a caseliner conforming to MIL-L-10547, type IV, grade E or F. When skids and reinforcement battens are required, the battens shall be placed on the box exterior and the skids shall be placed on the outside on the box bottom, in alignment with each side reinforcement batten. Box assembly, closure and strapping shall be as specified in appendix to the box specification except that the strapping shall be flat and the finish B. A groove shall be cut into the bottom face of each skid, full length at the center line, so that strapping, normally positioned adjacent to the skids, shall completely encompass the reinforcement battens and skids. Alternatively, when specified (see 6.2), the shipping container may be a close-fitting box conforming to PPP-B-601, overseas type, style A or B, with case liner as specified herein and with box lid secured to the box body with formed spring clips/fasteners and strapping in lieu of nailing and strapping. Blocking, bracing and cushioning shall be used in accordance with MIL-STD-1186.

5.2.2 Level B. Each complete drum, separate components and its technical manual shall be packed as follows. Size I, II and IIA drums shall be packed as specified in 5.2.1 except the box shall be domestic type and strapping shall be finish A. Size III drums shall be individually packed in a close-fitting box

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conforming to PPP-B-636, V3c, style FOL-L with glued manufacturer's joint. Box closure shall be method III and reinforcing strapping shall be nonmetallic in accordance with appendix to PPP-B-636. Blocking, bracing and cushioning shall be used in accordance with MIL-STD-1186.

5.2.3 Level C. Each complete drum, separate components and its technical manual shall be packed in close-fitting boxes conforming to either PPP-B-636, class domestic, or PPP-B-640, class 1, style and grade as appropriate. All items shall be blocked tightly to prevent movement and damage within the container. Box closure and reinforcement shall be in accordance with the appendix to the appropriate box specification.

5.2.4 Marking. Marking for military levels of protection shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The drums are intended for storage and transporting of drinking water. The drums should not be used for nondrinkable water or for fuel.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Size of drum required (see 1.2).
c. When a first article is not required for inspection and approval (see 3.2).
d. When certificate of compliance is required (see 3.4.1.1 and 3.4.11).
e. Degree of preservation and level of packing required (see 5.1 and 5.2).
f. When alternate PPP-B-601, style A or B box with special closure is required for level A (see 5.2.1).
g. When accessories for 250 gallon drum are not required (see 3.12).

6.3 First article. When a first article inspection is required, the items should be samples selected from the first 10 production items of each size. The first article should consist of 2 units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each drum.

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6.5 Subject term (key word) listing.

Blivet
Drinking water
Drum
Elastomer
Rubber
Water
Water drum

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - SA
Air Force - 69

Review activities:
Army - MD, GL
Air Force - 99, 43
DLA - GS

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 8110-0286

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-43699C	Drums, Fabric-Reinforced, Collapsible: (Drinking Water; 55-, 250- and 500 Gallon Capacities)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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